

Instructions (to students):

Run the following script to create three tables in your DB (MySQL/PostgreSQL-style):

```
CREATE TABLE students (  
    student_id INT PRIMARY KEY,  
    name VARCHAR(50),  
    dept VARCHAR(30),  
    batch INT,  
    cgpa DECIMAL(3,2)  
);  
  
CREATE TABLE courses (  
    course_id INT PRIMARY KEY,  
    course_name VARCHAR(50),  
    credits INT,  
    semester INT  
);  
  
CREATE TABLE enrollments (  
    enrollment_id INT PRIMARY KEY,  
    student_id INT,  
    course_id INT,  
    grade CHAR(2),  
    FOREIGN KEY (student_id) REFERENCES students(student_id),  
    FOREIGN KEY (course_id) REFERENCES courses(course_id)  
);
```

Insert sample data (upto 15 rows in a separate file).

Task:

List all students who are enrolled in at least one course, showing:

student_id, name, course_id, course_name.

Task:

Find all students (even those not enrolled in any course), showing their name, dept, and the course_name they are enrolled in; if not enrolled, show NULL for course_name.

Task:

list all students who are **not enrolled in any course** (student_id, name, dept).

Task:

Display each course (course_id, course_name) along with the **number of students enrolled** in that course. Include courses with zero enrollments.

Task:

Find students who have enrolled in **more than one course**, showing student_id, name, and total number of enrollments.

Task:

For each course (course_id, course_name), find the **student who got the highest grade** (assume grade is a string like 'A', 'B', 'C', etc.; you can treat alphabetically).

Show: course_id, course_name, student_id, name, grade.

Task:

Using a **self-join** on the students table, list pairs of students who belong to the **same department** but have different student_ids. Show:

s1.name AS student1, s2.name AS student2, s1.dept.

(Keep sample size small so pairs don't explode.)

Task:

Find all students (with dept = 'CSE') who have taken any course in semester = 5 and have a grade of 'A' or 'A+'. Show:

student_id, name, course_name, grade.

Task:

Using a RIGHT JOIN, write a query that returns all courses that have **no enrollments** (course_id, course_name, credits, semester).